

Sunlytics Project Proposal

A Data-Driven Framework for Solar Energy Readiness and Impact

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Executive Summary

Sunlytics is a data-driven analytics project that aims to evaluate how countries worldwide are progressing in their transition toward solar energy. By integrating multi-source datasets on solar capacity, environmental performance, and socio-economic indicators, Sunlytics builds a **Global Solar Transition Index (GSTI)** — a comprehensive measure that reflects each country’s advancement toward sustainable solar energy adoption. The project provides decision-ready insights for policymakers, investors, and researchers.

Project Overview

Goal. Develop a robust data platform and analytical capability to measure and compare country-level progress in adopting solar energy, producing an index (GSTI) that captures environmental, economic, social, and technical dimensions.

Scope. Data integration, cleaning and standardization; Data Warehouse implementation (star schema) on SQL Server; index computation; dashboards and visual analytics in Power BI and Tableau; final reporting and recommendations.

Objectives

1. Design and implement a centralized Data Warehouse that integrates solar-related indicators from multiple reliable sources.
2. Define and compute the Global Solar Transition Index (GSTI) combining relevant KPIs.

3. Identify leading and lagging countries in the solar transition and produce comparative analysis.
4. Deliver interactive dashboards and a final report to support policy and investment decisions.

Tools and Technologies

Python	Data cleaning, merging, preprocessing and automation scripts.
Microsoft Excel	Initial exploration, quick transforms and small-scale data checks.
SQL Server	Data Warehouse design and storage (star schema).
Power BI	Interactive dashboards and KPI visualizations for stakeholders.
Tableau	Advanced visual analytics and comparative reporting.

Methodology and Work Plan

1. Data Collection: gather datasets on solar capacity, generation, solar resource (GHI), LCOE, CO₂ emissions, GDP per capita, electricity access, tariffs, etc.
2. Data Cleaning & Transformation: use Python to standardize country names, harmonize units, remove duplicates and handle missing values.
3. Data Integration: merge cleaned sources into canonical tables and export to staging files for DWH load.
4. Data Warehouse Design: implement star schema in SQL Server.
5. Index Computation: define weights and normalization rules, compute GSTI score per country-year.
6. Visualization & Reporting: build dashboards in Power BI and Tableau; produce final report with policy recommendations.

Work Plan (high level)

Phase	Description	Duration
Data Collection	Acquire datasets from public databases and CSV/Excel sources	Week 1
Data Cleaning	Standardize names, handle nulls, unit conversions (Python)	Week 2
Data Integration	Merge sources into canonical tables and staging files	Week 3
DWH Design	Create star schema and tables on SQL Server	Week 4
Visualization	Build dashboards in Power BI and Tableau	Week 5
Reporting	Final report and presentation	Week 6

Stakeholder Analysis

Stakeholder	Role / Description	Benefit / Use-case
Government agencies	Ministries of Energy, Environment, Planning	Evaluate national progress; design policies and targets
Private sector	Solar companies, utilities, investors	Identify investment markets and technology opportunities
Researchers & academia	Universities and think tanks	Source of high-quality metrics for studies
International orgs	UN, World Bank, IRENA	Monitor progress, shape funding and programs

Expected Outcomes

- Centralized Data Warehouse containing harmonized solar transition data.
- Global Solar Transition Index (GSTI) scored for each country-year.
- Interactive dashboards (Power BI & Tableau) for stakeholders.
- Comprehensive report with findings, country-level profiles and policy recommendations.

Appendix: Key Data Attributes for the DWH

- **Dimensions:** dimCountry, dimCountryIndicators, dimEnvironment, dimEfficiency, dimGlobalSolarTrends.
- **Fact:** fact_GSTI (CountryID, Year, SolarCapacity_GW, SolarGrowth_TWh, CO2_perCapita, GDP_perCapita, LCOE, AccessToElectricity%, ReliabilityIndex, ...).

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